Project Report Paper on-
Emergency Medical Service Dispatch & Telematics Platform

Submitted to:

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CERTIFICATE

This is to certify that the Project Based Learning report entitled “Emergency Medical Service Dispatch & Telematics Platform” has been prepared by Rahul Mandal, Niraj Mandal, Manish Sah, Pushkar Priyadarshi, and Ashutosh Kumar of the Department of Information Technology, Arya College of Engineering & I.T., under the guidance of Er. Ram Babu Buri, Associate Professor, for academic submission to Rajasthan Technical University, Kota.

The report presents the project concept, system design, major modules, algorithms, security approach, testing strategy, limitations, and future scope of the H8 capability-aware emergency medical dispatch and telematics platform.

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DECLARATION

We hereby declare that this Project Based Learning report titled “Emergency Medical Service Dispatch & Telematics Platform” has been prepared for academic submission by the undersigned students. The report has been organized from the project engineering dossier and the project-report material supplied by the team.

The technical architecture, algorithms, performance targets, security controls, and operational practices described in the report are presented as project design and engineering specifications unless an implementation result is explicitly identified. No unverified production or clinical outcome is claimed.

Student signatures: _____

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We acknowledge all team members for their contributions to problem analysis, system design, algorithmic modelling, documentation, testing, and preparation of this report.

ABSTRACT

Emergency medical response is a time-sensitive coordination problem in which selecting the nearest ambulance is not always sufficient. A nearby unit may lack the clinical capability required by an incident, have stale location information, or be needed to preserve coverage elsewhere. Similarly, hospital selection should consider capability, capacity, diversion status, and expected travel time rather than distance alone.

The Emergency Medical Service Dispatch & Telematics Platform, referred to as H8 in the engineering dossier, is designed as a distributed decision-support system for metropolitan emergency response. The project uses a Jaipur pilot context with 14 ambulance units and five Tier-1 receiving hospitals. It combines incident intake and triage, capability-aware ambulance ranking, safe unit reservation, GPS telemetry, road-network routing, hospital destination ranking, fleet redeployment, and an auditable event trail.

The technical foundation uses Java 21, Spring Boot, PostgreSQL/PostGIS, Apache Kafka, Redis, GraphHopper, Keycloak, and a progressive web application. The report focuses on the project design and engineering approach rather than presenting the dossier's benchmark targets as independently measured results.

Keywords: Emergency Medical Services, dispatch, telematics, ambulance allocation, geospatial routing, hospital selection, auditability, privacy.

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CHAPTER 1 – INTRODUCTION

1.1 Background

Emergency medical services require rapid coordination among callers, dispatchers, ambulance crews, routing systems, and receiving hospitals. A conventional closest-vehicle approach is useful as a starting point, but distance alone cannot represent clinical suitability, telemetry quality, remaining fleet coverage, or hospital readiness.

1.2 Project Overview

H8 is designed as a capability-aware dispatch and telematics platform. Its purpose is to support trained dispatch personnel with consistent, explainable resource recommendations while retaining human control over operational decisions.

1.3 Operational Context

Element	Project Context
Pilot geography	Jaipur, Rajasthan, India
Ambulance fleet	14 active units in the baseline scenario
Receiving facilities	5 Tier-1 hospitals
Primary users	Dispatchers, ambulance crews, hospital staff, auditors
Core response loop	Incident → triage → dispatch → tracking → hospital handover → audit

1.4 Need for the Project

- Capability mismatch can occur when a nearby ambulance is not equipped for the incident.
- Hospital-blind transport can increase diversion, delay, or secondary transfers.
- Stale telemetry can make a nominally nearest unit an unreliable choice.
- Dispatching one unit can reduce coverage in another area.
- Fragmented records make review and accountability more difficult.

CHAPTER 2 – PROBLEM DEFINITION AND OBJECTIVES

2.1 Problem Statement

The project addresses the limitations of distance-first emergency dispatch. The system must balance clinical suitability, estimated travel time, unit availability, location freshness, and the effect of a dispatch on remaining fleet coverage. It must also support destination decisions that account for hospital readiness.

2.2 Objectives

- Represent incident urgency and clinical need in a structured form.
- Rank eligible ambulances using multiple operational and clinical factors.
- Prevent double booking through atomic reservation.
- Provide live unit location and route ETA using current telemetry.
- Rank receiving hospitals using capability and operational readiness.
- Support fleet redeployment to improve expected coverage.
- Maintain a tamper-evident record of important decisions and overrides.
- Provide secure, role-based access to operational functions.

2.3 Scope

The project covers the software workflow from incident intake through dispatch, tracking, destination selection, handover, and audit. It includes the supporting data, messaging, geospatial, security, and operational design needed for this workflow.

2.4 Users and Responsibilities

Role	**Primary responsibility**
Dispatcher	Review incidents, dispatch recommendations, overrides, and workflow status.
Crew	Receive assignments, transmit telemetry, and update operational status.
Hospital Staff	Provide receiving-facility readiness and handover information.
Auditor	Review decision history and verify audit-chain integrity.

CHAPTER 3 – PROPOSED SYSTEM AND METHODOLOGY

3.1 Development Approach

The project follows a modular, event-driven software architecture. Domain decisions are separated from infrastructure so that dispatch and coverage logic can be tested independently. Services communicate through synchronous APIs where immediate responses are needed and asynchronous events where state changes can be propagated reliably.

3.2 End-to-End Workflow

• Incident is received and identifying caller information is minimized or anonymized.

• Incident severity and clinical need are recorded.

• Eligible ambulance candidates are retrieved.

• Candidates are scored using ETA, capability, coverage, fatigue and telemetry freshness.

• The selected unit is atomically reserved to prevent double assignment.

• The crew receives the assignment and telemetry updates begin.

• Destination hospitals are ranked using capability, capacity and diversion information.

• Pre-arrival and handover events are recorded.

• Important actions and overrides are added to the audit chain.

3.3 Design Principles

- Single algorithmic truth for core dispatch and coverage decisions.
- Human dispatcher remains responsible for operational overrides.
- Stale data is explicitly penalized rather than silently trusted.
- Event processing is idempotent to tolerate duplicate messages.
- Sensitive caller information is minimized and protected.
- Operational decisions remain traceable for later review.

CHAPTER 4 – SYSTEM ARCHITECTURE AND MAJOR MODULES

4.1 Architecture

The platform is organized as cooperating services around a common domain model. The gateway provides controlled access to the platform, while specialized services manage incidents, dispatch, tracking, routing, hospitals, redeployment, and auditing.

Service	Main responsibility	Port
Gateway	Entry point, authentication and routing	8080
Incident	Incident intake and lifecycle	8081
Dispatch	Candidate ranking and reservation	8082
Tracking	GPS telemetry and unit state	8083
Routing	ETA and route matrix	8084
Hospital	Destination ranking and handover	8085
Redeployment	Coverage analysis and repositioning	8086
Audit	Tamper-evident event ledger	8087
Web/Proxy	Unified web access	8088

4.2 Technology Stack

The engineering dossier specifies Java 21 LTS and Spring Boot for services, PostgreSQL 15 with PostGIS for persistent and geospatial data, Apache Kafka for event streaming, Redis for fast geospatial/state access, GraphHopper for road routing, Keycloak for OAuth2/OIDC and role-based access, and a PWA-oriented frontend.

4.3 Architectural Benefits

- Separation of responsibilities
- Scalable event-driven communication
- Independent testing of domain logic
- Controlled access through a gateway and identity service
- Geospatial support for live fleet and route calculations

CHAPTER 5 – SYSTEM DESIGN AND WORKING

5.1 Incident Management

An incident moves through a controlled lifecycle: REPORTED, TRIAGED, DISPATCHED, ON_SCENE, TRANSPORTING and RESOLVED, with cancellation available where appropriate. Severity classes include CRITICAL, URGENT, STANDARD and NON_EMERGENCY. Clinical needs include CARDIAC, TRAUMA, STROKE, PEDIATRIC, RESPIRATORY and GENERAL.

5.2 Ambulance Dispatch

The dispatch service first identifies eligible units and then evaluates them using the common scoring model. A capability mismatch is penalized when the required level of care is not available. The final recommendation is therefore based on more than geographic proximity.

5.3 Live Tracking and Routing

Ambulance telemetry can be ingested at up to 1 Hz per unit in the design. Redis geospatial state uses a 60-second freshness window. When recent GPS data is temporarily unavailable, dead reckoning may be used; data older than the defined freshness limit is treated as stale.

GraphHopper is the primary road-network routing mechanism. The engineering design also specifies a fallback based on Haversine distance with a 1.35 tortuosity factor and 40 km/h assumed speed when routing fails or exceeds the defined timeout.

5.4 Hospital Destination

Hospital selection considers emergency capability, current capacity or diversion status, travel time, and other operational factors. A pre-arrival event can be sent so that the receiving facility can prepare before the ambulance arrives.

CHAPTER 6 – ALGORITHMS AND DATA FLOW

6.1 Capability-Aware Dispatch Scoring

The dispatch score is a multi-factor decision model rather than a simple nearest-distance rule. Candidate quality is influenced by ETA, clinical capability, fleet coverage effect, crew fatigue, and telemetry freshness. The purpose is to select a unit that is both useful for the incident and safe for the wider fleet.

6.2 Coverage Preservation

The design uses a Maximum Expected Covering Location Problem (MEXCLP)-based approach to estimate the effect of removing a unit from an area. An eight-minute coverage threshold is used in the source design. If dispatching a unit creates an important coverage gap, redeployment recommendations can be generated.

6.3 Atomic Reservation

After ranking, the selected unit must be reserved atomically. The conditional database update ensures that two dispatchers or concurrent requests cannot successfully reserve the same unit. This is a key correctness property of the system.

6.4 Event and Data Flow

Stage	Important data	Outcome
Intake	Severity, clinical need, location	Structured incident
Candidate search	Unit state, capability, telemetry	Eligible set
Scoring	ETA, capability, coverage, freshness	Ranked candidates
Reservation	Unit status/version	Single confirmed assignment
Tracking	GPS, timestamp, status	Live operational view
Destination	Hospital capability/capacity	Recommended facility
Handover	Arrival and transfer details	Completed incident
Audit	Action + hash chain	Verifiable history

CHAPTER 7 – SECURITY AND PRIVACY

7.1 Authentication and Authorization

The design uses Keycloak with OAuth2/OIDC and role-based access. Dispatcher, crew, hospital staff and auditor functions are separated so that users receive only the permissions needed for their operational role.

7.2 Data Protection

- TLS 1.3 is specified for protected network communication.
- Caller phone information is represented through salted SHA-256 hashing rather than plaintext storage.
- Secrets must remain outside source control.
- Operational actions and overrides are recorded for accountability.
- Audit records are linked using SHA-256 hashes to make tampering detectable.

7.3 Safety-Oriented Controls

Because dispatch is safety-sensitive, the design does not assume that automation is infallible. Manual override, stale-data visibility, routing fallback, safe reservation semantics, and auditable decisions are treated as operational safeguards.

7.4 Privacy and Governance

The project design minimizes unnecessary personally identifying information. Before real deployment, the clinical rules, privacy practices, data retention, regulatory obligations, and organizational approvals must be reviewed by the responsible authorities.

CHAPTER 8 – TESTING AND EVALUATION APPROACH

8.1 Verification Strategy

The engineering dossier identifies unit testing, property-based testing, concurrency testing, integration testing, performance testing, resilience testing, security testing, and usability evaluation. These categories are retained here as the project's verification approach.

8.2 Important Test Areas

Area	Example verification
Dispatch logic	Correct ranking under different clinical and ETA conditions
Reservation	Concurrent requests cannot reserve one unit twice
Telemetry	Fresh, delayed and stale GPS states behave correctly
Routing	Primary routing and fallback calculations are handled safely
Hospital ranking	Capability and diversion information affect selection
Messaging	Duplicate events do not produce duplicate state changes
Audit	Hash-chain verification detects altered records
Security	Unauthorized roles cannot perform restricted actions

8.3 Performance Targets

The source dossier specifies design targets including dispatch scoring below 50 ms P95 for 50 active ambulances, telemetry ingestion of at least 1,000 events per second with low response latency, gateway overhead below 5 ms, and a target availability of 99.99%. These figures are acceptance targets from the engineering source, not independent measurements reported by this academic report.

8.4 Evaluation Principle

Final implementation testing should record the test environment, input scenario, expected result, observed result, and pass/fail status. No benchmark is presented as an achieved result unless it has been actually measured.

CHAPTER 9 – LIMITATIONS AND FUTURE SCOPE

9.1 Limitations

- Real travel time depends on traffic, road conditions, weather and local operating conditions.
- Clinical capability rules require validation by qualified EMS and medical stakeholders.
- Hospital capacity information is useful only when updated accurately and promptly.
- GPS and network failures can affect ETA and ranking.
- Benchmark results depend on the hardware, data and workload used for testing.
- Production deployment requires formal security, privacy, regulatory and operational review.

9.2 Future Scope

- Demand forecasting and proactive fleet positioning.
- Validated patient and vehicle telemetry integration.
- Traffic-aware routing and real-time road-condition data.
- Multilingual and accessibility-focused interfaces.
- Cross-city deployment and disaster-recovery expansion.
- Simulation tools for staffing, coverage and policy evaluation.
- Operational analytics for response time, offload and fleet utilization.

9.3 Expected Project Value

The principal value of the proposed system is a consistent decision-support workflow that considers clinical capability and operational context together. It is intended to reduce avoidable capability mismatches and improve visibility of fleet and hospital status while preserving human oversight.

CHAPTER 10 – CONCLUSION

The Emergency Medical Service Dispatch & Telematics Platform presents a structured approach to emergency-response coordination. Instead of treating proximity as the only dispatch criterion, the system combines clinical need, ambulance capability, road-network ETA, telemetry freshness, fleet coverage, and hospital readiness.

The project brings these functions into a coordinated architecture using modular services, geospatial data, event-driven communication, secure access control, and an auditable decision trail. Atomic reservation and idempotent processing address important concurrency and reliability concerns, while stale-data handling and routing fallback improve operational resilience.

For academic purposes, the platform demonstrates how software engineering, algorithms, databases, distributed systems, geospatial processing, security, and human-centred operational design can be combined in one PBL project. Before real-world use, however, the design must be implemented, tested with representative data, clinically validated, security reviewed, and approved by the relevant organizations.

Key Outcome

The project establishes a professional software architecture and decision-support model for capability-aware EMS dispatch, with clear boundaries between design specifications and independently verified implementation results.

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[4] PostgreSQL Documentation, PostgreSQL 15.

[5] PostGIS Documentation, PostGIS 3.4.

[6] Apache Kafka Documentation, Kafka 3.7.

[7] Keycloak Documentation, Keycloak 25.

[8] GraphHopper Documentation, routing and travel-time services.

[9] OWASP Application Security Verification Standard and secure web application guidance.

APPENDIX – SELECTED PROJECT ARTEFACTS

The final submission may include only the diagrams and screenshots that correspond to the implemented project. Recommended figures are the system architecture, use-case diagram, dispatch workflow, ambulance state model, and deployment overview. Unimplemented screens or unsupported results should not be presented as completed features.

This report intentionally excludes the lengthy SRS-style appendices, full API catalogue, operational runbooks, and source-dossier benchmark tables because they are not necessary for a concise PBL project report.